

Problem Statement ver. 8/10/23

As a team, complete the assignment outlined below. Include a cover sheet with the assignment submittal and include a brief executive summary. The cover sheet for any team-based senior design submittal should include the name of the submittal, the name of the project, the date, and the name(s) of all participants.

Part A: Problem Statement

First, define the problem. The problem statement is one sentence (ideally) that clearly and concisely defines the problem to be solved. It is stated in the language of the customer or end user of the project item. It might be necessary to provide additional paragraphs that provide context to the problem. This helps define the scope and/or the customer or end-user.

Part B: Requirements and Creating Value

Describe requirements the customer or end-user would expect of the design project. These should be high-level requirements that the design project should satisfy. Describe how this design will create value for the customer or end-user.

Part C: Existing Solutions

Next, include a section on existing design solutions you have found. What part(s) of your problem do they solve? What aspects or parts of the problem are not satisfied? What are the requirements and specifications of these existing solutions, if available? What are their costs? What are their shortcomings? Do they comply with any engineering standards (if so, identify the standards with which they comply)?

Carefully consider how you organize this information. How should you best communicate your findings? What format(s) should you use? Tables are great ways to provide information for comparison.

Part D: Solution

Finally, describe your team's initial thoughts and possible approach to providing a solution to the problem. Identify the benefits offered by your intended solution, how your solution will create value, and how it differs from any existing solutions.

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Problem Statement Grading

Category	Below	Adequate	Above	Comments
Real problem identified	☐	☐	☐	_____
Clearly defines problem	☐	☐	☐	_____
Concisely defines problem	☐	☐	☐	_____
Stated in the customer's language	☐	☐	☐	_____
Design project requirements identified	☐	☐	☐	_____
Envisioned solution and its benefits	☐	☐	☐	_____
Existing solutions: How solved	☐	☐	☐	_____
Existing solutions: Requirements/Specs	☐	☐	☐	_____
Proposed Solution: Benefits	☐	☐	☐	_____
Proposed Solution: Value to customer	☐	☐	☐	_____
Writing quality	☐	☐	☐	_____

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Problem Statement

Smart Home Manager

09/13/26

Miles Bailey-Waltz

Jackson Duty

Megan Ruffing

Lily Schellenbach

Atifa Tariq

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Part A: Problem Statement

In the modern age, many problems of humanity have been automated away. However, these automations typically come with numerous individual sensors that all need their own integrations. Matter is an open source platform that fixes this integration program by allowing anyone to contribute to a growing library of smart devices by using Matter TCP/IP. However, each of these devices typically need its own sensors to function properly. This can get more cumbersome the more devices are added, and can lead to headaches. To address this, our team will be designing a set of small, affordable, and wireless sensors to allow greater control due to their versatility. These devices will be compatible with Matter, which will make them compatible with existing hubs from Google, Apple, and Amazon.

Part B: Requirements and Creating Value

The target of this product is anyone currently inside the matter ecosystem. This means anyone with a smart home hub that is compatible with Matter TCP/IP. Customers would expect the system to accurately monitor room conditions such as ambient temperature and light levels, and occupancy. These sensors should communicate wirelessly with the Matter hub while being cheap to produce and energy efficient enough to be powered for a full week by 2 AA batteries. Our design creates value by enhancing existing home automation systems, sending reliable data to control many smart devices within the ecosystem.

Part C: Existing Solutions

Matter has existing sensors, but because it was designed to integrate existing smart home applications, most of its existing sensors are expensive for their application because they are designed to be stand alone systems that integrate into Matter. If a person who already has a thermostat wants to measure the temperature downstairs using Matter, the cheapest standalone temperature sensor is the Switchbot Room Thermometer, which retails on amazon for \$13.99(https://www.amazon.com/dp/B07L4QNZVF?ref=emc_p_m_5_i_atc&th=1). This is a combination temperature and humidity sensor, and while it is inexpensive, it uses an LCD screen. This means that there is room for the device to be smaller and more inexpensive than it currently is. Switchbot is the least expensive standalone temperature sensor by a wide margin, with the next best Matter compatible option in the United States being the Aqara Climate Sensor W100(<https://www.amazon.com/dp/B0DWSK5H33?tag=matteralpha-20&linkCode=osi&th=1&psc=1>) which retails for \$37.99. The pickings for light sensors are similar, but lack a Switchbot design. The minimum price is Aquara's motion and light sensor P2, which retails for \$33.99 on their website, but sells for \$52 at other retailers.(https://us.aqara.com/products/motion-and-light-sensor-p2?variant=49161688711450&country=US¤cy=USD&utm_medium=product_sync&utm_source=google&utm_content=sag_organic&utm_campaign=sag_organic&srsId=AU7gw4V2NnKLvjyfQYjqDa45oNpxudpBYUCqPDCMS6---mEmv0uS_1fUwk),(https://www.walmart.com/ip/Aqara-Motion-and-Light-Sensor-P2-White-ML-S03D/15382812367?wmlspartner=wlp&selectedSellerId=1330&veh=seo_fpl&cn=google).

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Part D: Solution

Our solution to this is to design several small, distributed boards that run wirelessly off AA batteries. This will allow the end user to place them at their leisure and will seamlessly integrate into the existing Matter framework so that home automation becomes simple. These boards will communicate over Matter's existing Wi-Fi protocol to send data to the hub and will be designed to be as minimalist as possible. The designs will be open source, and the data will be exclusively shared with the Matter hub.